

Stem Cells: Evidence Before Judgement

I can describe stem-cell functions and distinguish scientific evidence, ethical concern and uncertainty.

FUNCTION

divide and differentiate

EVIDENCE

what observations support

ETHICS

values as well as facts

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What success looks like

- I can describe a stem cell as able to divide and differentiate.
- I can distinguish embryonic stem cells, adult stem cells and plant meristems.
- I can classify a statement as evidence, ethical concern or uncertainty.

Retrieve and reconnect

- State one role of mitosis in growth or repair.
- Describe differentiation in one sentence.
- Name the plant tissue that contains stem cells.
- Lead-in: why is 'might help' different from 'has been shown to help'?

A headline says, 'Stem cells will cure every damaged organ.' What should a scientist do first?

Think first. Point, say, sign or write your choice. Give one reason.

- A. Accept it because stem cells can differentiate.
- B. Check the evidence, the cell type, the outcomes and the risks.
- C. Reject every possible medical use of stem cells.

Look closely · what is the evidence?

CLAIM · what is being said

EVIDENCE · what was observed

LIMIT · what is still uncertain

Look closely · what is the evidence?

CLAIM

"Stem cells could treat this condition."

EVIDENCE

In a lab, cells differentiated into the needed type

LIMIT

no patient trial yet · small sample · source of cells raises questions

Claim

Evidence

Limit

Words we will use

- stem cell — unspecialised starting cell
- differentiate — become specialised
- ethical — about what ought to be done

Watch the model

- Begin with function: stem cells can divide, then some daughter cells can differentiate into specialised cells.
- Distinguish sources: embryonic stem cells can form many cell types; adult stem cells usually form a more limited range; meristems produce new plant cells.
- Keep claim and evidence separate: a biological ability creates a possible use, but clinical benefit and safety still require evidence.

Which statement describes a stem-cell function rather than only a property?

- A. It can divide and produce cells that differentiate, supporting growth or replacement.
- B. It is called undifferentiated.
- C. It cures every disease.

Commit to an answer. Show the evidence you used. Listen, then revise if needed.

Build the explanation

- Scientific evidence can estimate possible benefit and harm; it cannot decide a moral value by itself.
- An ethical concern may focus on the source of cells, consent, fairness or acceptable harm.
- A careful judgement names what is known, what is valued and what remains uncertain.

Find and repair the misconception

If stem cells can become specialised cells, a stem-cell treatment is automatically safe and effective.

A. Keep it: biological possibility proves clinical success.

B. Repair it: possible use still needs evidence of benefit, risk and long-term outcome.

C. Repair it: no stem-cell research can ever be ethical.

Evidence, ethics or uncertainty?

Place each statement by its main role. Then explain why one statement could matter to more than one column.

- READ
- CLASSIFY
- JUSTIFY
- CHECK
- REPAIR

Evidence, ethics or uncertainty?

Use the numbered cards in your pupil pack. Take turns to choose, justify and check. You may direct a partner to move a card.

Choose a group · defend your placement

- SCIENTIFIC EVIDENCE
- ETHICAL CONCERN
- UNCERTAINTY

Which column contains a fact, which contains a value question and which names a gap in evidence?

Your independent mission

Build a stem-cell evidence glossary and classify three short statements before writing one cautious scientific claim.

Record: An accurate glossary, classified statements and one cautious claim that distinguishes possibility from evidence.

Choose your support and challenge

- Supported: Match stem cell, differentiate and ethical to icon-backed meanings; sort three statements.
- Standard: Classify three statements and describe one stem-cell function before writing one cautious claim.
- Stretch: Classify three statements, identify one overlap and explain why function alone cannot prove treatment success.

Show what you know

Distinguish scientific evidence from an ethical concern in this context.

Point to, read or explain the evidence you made. Choose one next step after the adult has listened.